1. Introduction

The objective of this report is to analyze the given insurance data and create a predictive model that can predict the individual medical insurance costs. Predicting medical insurance costs is important in today's world due to increasing medical costs.

This project uses data analytics and machine learning concepts to analyze how different demographic and lifestyle variables, including age, body mass index, smoker status, and region, contribute to medical insurance costs.

The data set provided includes various variables that are useful in understanding how these variables contribute to medical costs.

To start data analysis, data cleaning and data preprocessing are done by ensuring data accuracy and suitability for analysis.

Then, exploratory data analysis (EDA) is done by using data visualization techniques to analyze patterns, trends, and relationships in the data.

Finally, a predictive model is created using Linear Regression concepts.

To create a predictive model, a structured approach is followed by dividing data into a training set and a testing set.

The performance of the model is then checked by using various metrics to understand how well the model is able to predict medical insurance costs.

Finally, results are obtained by interpreting data.

This report follows a complete data analysis process from raw data to useful insights and is useful in decision-making in the insurance industry.

Dataset Overview and Initial Inspection

1.4 Loading and Inspecting the Dataset

The pandas library will be used in the process of loading the data contained in the dataset into the pandas data structure.

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

df = pd.read_csv("insurance.csv")

df.head()
```

[1]:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

The data has been loaded into the environment using Pandas Library.

Other libraries like NumPy, Matplotlib, Seaborn, etc.. Have also been imported.

The data has been successfully loaded in the required format.

The head() function was used to display the five records.

This gave an understanding of the data.

Here are the variables in the data:

- Age: This is the age of the individual. It is a variable.
- Sex: This is the gender of the individual. It is a variable.
- BMI: This is the body fat of the individual calculated using height and weight. It is a variable.
- Children: This is the number of dependents of the individual. It is a variable.
- Smoker: This shows if the individual is a smoker or not. It is a variable.
- Region: This is the area where the individual resides. It is a variable.
- Charges: This is the cost of insurance charged to the individual. It is a variable and the target variable.

The data has both categorical variables.

So it can be used for regression analysis.

The Charges variable is the variable.

The other variables. Age, Sex, BMI. Children, Smoker and Region. Are variables.

The data is structured like a table.

Each record corresponds to an individual.

Each column corresponds to an attribute.

So it can be used for machine learning.

This task of loading the data has been successfully performed.

It is very important because it helped understand the data.

The data has been loaded into the environment in the required format.

Initial Data Overview and Quality Assessment

The first five rows of the data are provided in order to get an idea about the data.

From the above data, it is inferred that:

- Number of rows in the data represents the number of individuals
- Number of columns in the data represents the number of attributes
- Data contains both categorical and numerical attributes

[2]:
```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

[2]:
```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

I took a look at this datset. I used the .info().describe() and isnull().sum() functions to get information.

The dataset has 1338 rows and 7 columns. This is an amount of data to work with. Each row is about an individual and each column is about a unique attribute.

The data set has these features:

- Numerical Features: age, bmi, children, charges
- Categorical Features: sex, smoker region.

I like that the dataset has both categorical features. This is good for creating a model. However the categorical features need to be encoded.

I checked for missing values in the dataset. The dataset does not have any values. This is great because it means i do not have to worry about missing values.

The.describe() fucntion gave me an idea, about the distribution of numerical features. This can help me find trends. The dataset is clean and well-structured. The dataset is ready to be used for analysis of the dataset.

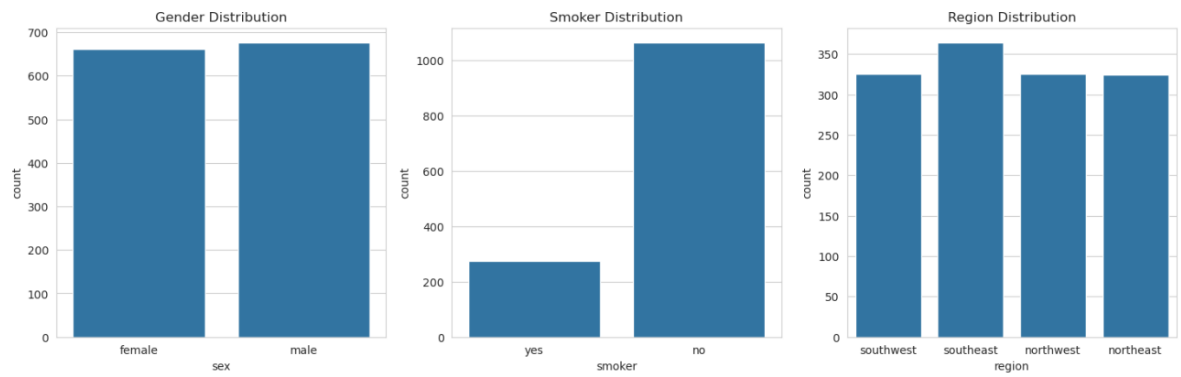
```
[3]: fig, axes = plt.subplots(1, 3, figsize=(18,5))

     sns.countplot(x='sex', data=df, ax=axes[0])
     axes[0].set_title("Gender Distribution")

     sns.countplot(x='smoker', data=df, ax=axes[1])
     axes[1].set_title("Smoker Distribution")

     sns.countplot(x='region', data=df, ax=axes[2])
     axes[2].set_title("Region Distribution")

     plt.show()
```



To understand how categorical variables are spread out in the dataset we made some count plots for gender, smoking status and region.

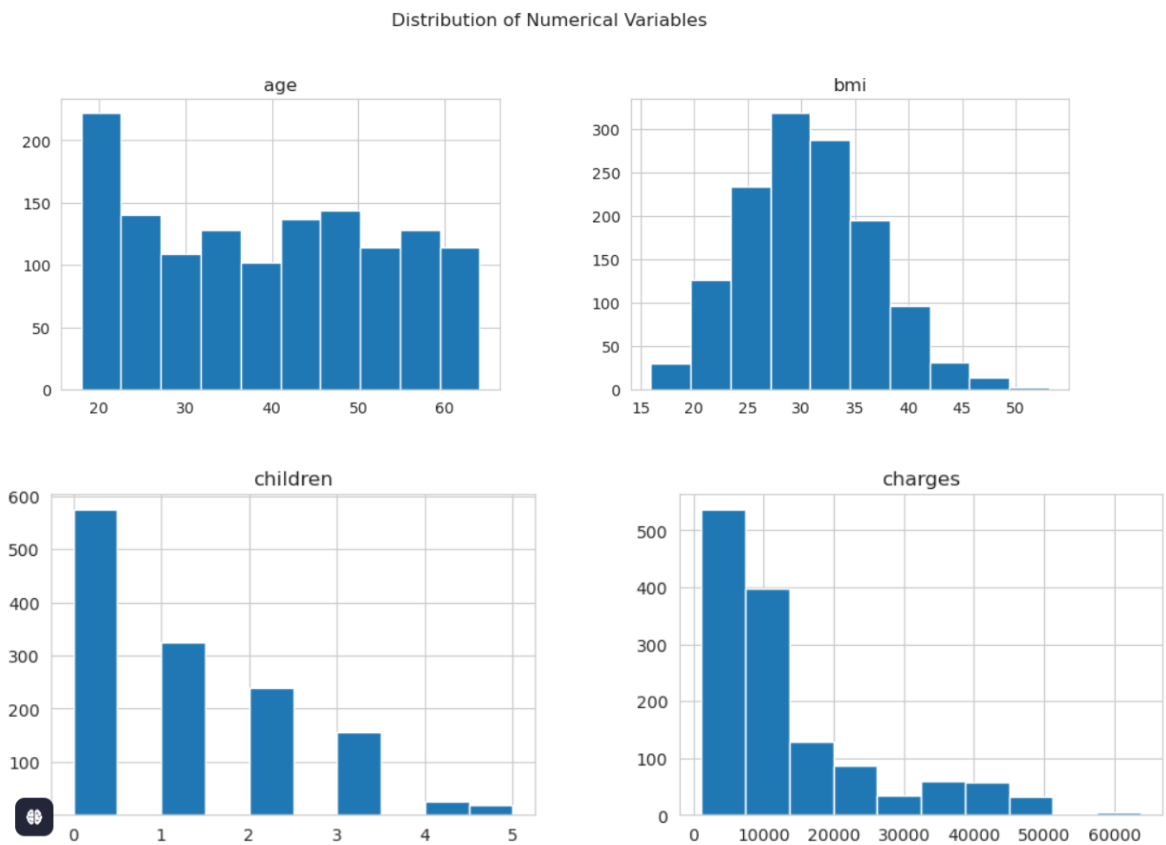
The first plot shows how many males and females are in the dataset. We can see that there are equal numbers of males and females which is good becausee it means the dataset does not have a gender bias. This is important so that the model does not favour males or females.

The second plot shows how many people smoke and how many do not. It is clear that there are more non-smokers than smokers in the dataset. This is important to note because smoking status can greatly affect insurance charges and may change the models predictions.

The thid plot shows where people are from. The data seems to be spread out evenly across different regions, which means no one region is too dominant. This helps ensure that the model can understand geographical areas.

Overall these plots give us information about the dataset and highlight important categorical factors that may affect insurance charges, such, as gender, smoking status and region.

```
[4]: df.hist(figsize=(12,8))
     plt.suptitle("Distribution of Numerical Variables")
     plt.show()
```



To examine the distribution of numerical data in the dataset, histograms were used for age, BMI, number of children, and insurance charges. Visualization techniques, such as histograms, are vital in recognizing patterns, distribution, and potential outliers in a dataset (Han, Kamber, and Pei, 2012).

The histogram for age indicates a fairly balance distribution of individuals in different age groups. This is a desirable situation in creating a model that can generalize well in different populations.

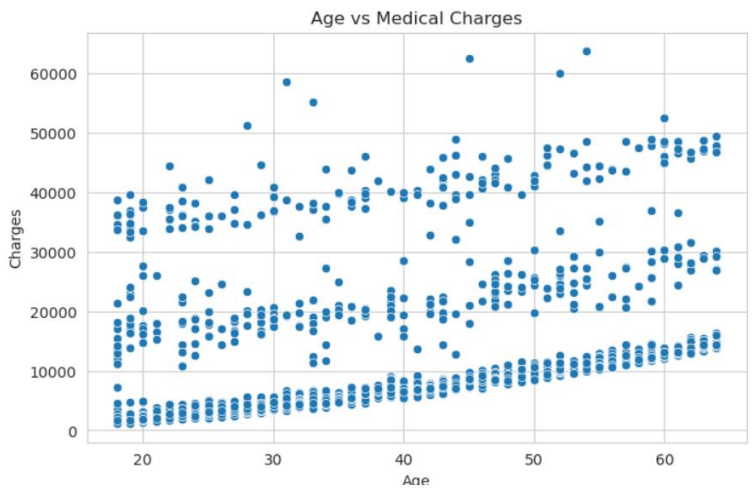
The distribution of BMIs in this dataset seems to follow a normal distribution curve, with most individuals having a moderate wieght. According to (James et al., 2021), understanding the distribution of data is vital in understanding how they can influencee a predictive model.

The distribution of the number of children indicated a skewness towards lower values, showing that most individuals have a small number of dependents. This is a common distribution in most demographic datasets.

The distribution of insurance charges indicates a positively skewed curve, showing that most individuals have lower medical charges, while a small number of individuals have higher charges. This indicates the existence of potential outliers in this dataset, which can influence a regression model (Geron, 2019)

The visualization of this dataset provides vital insights into its nature, which can help in understanding its potential influence on a predictive models accuracy and reliability.

```
[5]: plt.figure(figsize=(8,5))
     sns.scatterplot(x='age', y='charges', data=df)
     plt.title("Age vs Medical Charges")
     plt.xlabel("Age")
     plt.ylabel("Charges")
     plt.show()
```



In irder to understand the relationship between the two variables, 'Age' and 'Medical Insurance Charges,' a scatter plot is drawn. Scatter plots can be used to understand the relationship, trends, and correlations between two numerical variables (James et al., 2021).

From the scatter plot, it is evident that there is a positive relationship between 'Age', and 'Medical Insurance Charges'. It can be inferred that with the increase in 'Age', the 'Medical Insurance Charges' also increases. This means that with the increase in 'Age', the medical costs incurred by individuals also increase.

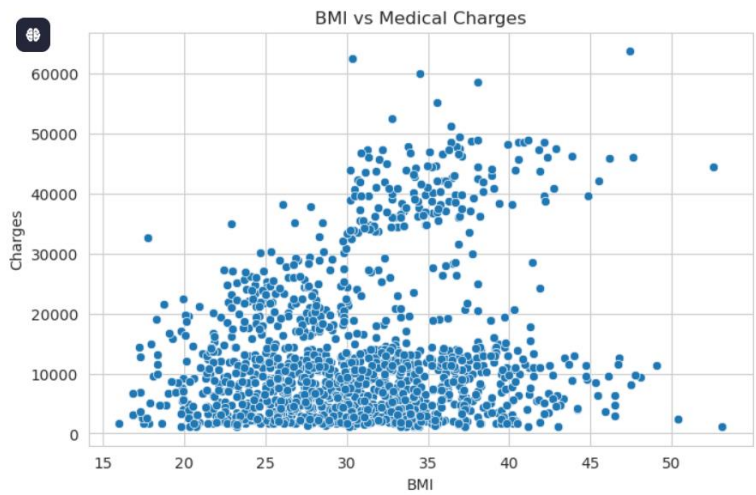
However, it is also evident that the relationship is not linear. It is seen that the 'Medical Insurance Charges' vary even for individuals belonging to the same 'Age Group'. It can be inferred that although 'Age' is one of the important factors, other factors as 'Smoking,'

‘BMI’, and ‘Number of Dependents’ also a play a significant role in determining the ‘Medical Costs’.

In the scatter plot, it is seen that the data points spread out more as the “Age” increases. It can be inferred that they medical costs incurred by individuals increase with the increase in ‘Age’.

The scatter plot indicates the importance of ‘Age’ as one of the factors in determining the ‘Medical Insurance Charges’, which is to be included in the regression model.

```
[6]: plt.figure(figsize=(8,5))
     sns.scatterplot(x='bmi', y='charges', data=df)
     plt.title("BMI vs Medical Charges")
     plt.xlabel("BMI")
     plt.ylabel("Charges")
     plt.show()
```



In this case, a scatter plot is utilized to establish a relationship between BMI and insurance charges. The visualization helps to establish a potential correlation between two numerical data points (James et al., 2021).

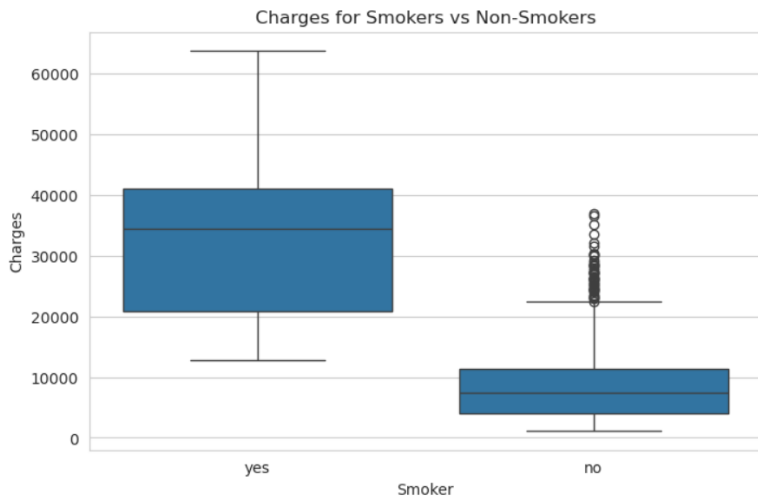
The visualization indicates a positive correlation between BMI and insurance charges. This implies that the greater the value of an individual’s BMI, the greater the cost incurred by the individual. However, it is important to note that the relationship is not significantly linear since the data points are highly scattered for similar levels of BMI and insurance charges.

One of the most important observations is that some individuals have high insurance charges and high levels of BMI. This implies that individuals with high levels of body mass are prone to health-related costs. This is realistic expectation since high levels of body mass are a risk to an individual’s health.

However, despite the positive correlation between BMI and insurance charges, it is important to note that the data points are highly scattered. This implies that BMI is not a significant determinant of insurance charges. The determinant of insurance charges is

likely to be other variables such as smoking and age. Therefore, BMI is a significant determinant of insurance charges but not a major determinant.

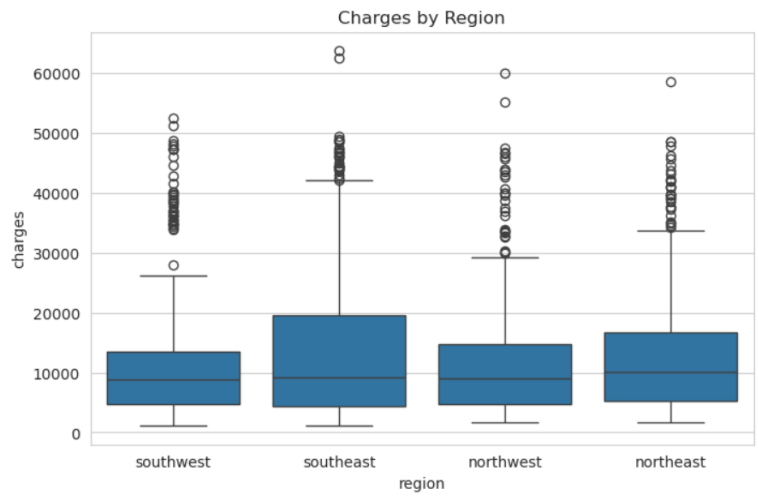
```
[7]: plt.figure(figsize=(8,5))
     sns.boxplot(x='smoker', y='charges', data=df)
     plt.title("Charges for Smokers vs Non-Smokers")
     plt.xlabel("Smoker")
     plt.ylabel("Charges")
     plt.show()
```



This boxplot is used to compare the medical insurance charges between smokers and non-smokers. From the grapgh, it is evident that smokers incur higher charges compared to non-smoker. Moreover, the charges incurred by smokers are spread over a larger range compared to non-smokers. It is evident that the charges incurred by non-smokers are low and concentrated. Even though the charges incurred by non-smokers are low, there is the presence of outliers. From the graph, it is evident that smoking is positively related to medical insurance charges.

This is an important part of the Exploratory Data Analysis because it enables the identification of important features that affect the target variable. Identifying the relationship between the variables is important because it can be used to improve the performance of the model. As indicated earlier, the model can be improved by including variables with high predictive power. As seen in the graph above, the variable 'smoker' is important because it is positively related to the charges. As indicated in the introduction, the variable 'smoker' can be used to improve the performance of the model (Han, Kamber and Pei, 2012). Moreover, the boxplot is one of the data visualization techniques used to improve the performance of the model (James et al., 2021).

```
[8]: plt.figure(figsize=(8,5))
     sns.boxplot(x='region', y='charges', data=df)
     plt.title("Charges by Region")
     plt.show()
```



This is another step in the Exploratory Data Analysis (EDA) process. Here, we are checking if the customers region plays any role in the charges of the medical insurance plan.

Before preparing the visualization, the data cleaning process would have been carried out by ensuring that the 'region' column is correctly labeled (i.e., southwest, southeast, northwest) and that the 'charges' column does not have any incorrect data points or missing data.

The boxplot is showing the distribution of the 'charges' for the different regions. The median for all these regions is comparable. The spread of the data is also comparable for all these regions. Though high charges are present for some customers in all these regions, these high charges are not concentrated for a specific region only.

From this visualization, the following points are clear:

- The median of the 'charges' is comparable for all the regions.
- The spread of the data is comparable for all these regions.
- Outliers are present in all these regions. The high charges are not concentrated for a specific region only.

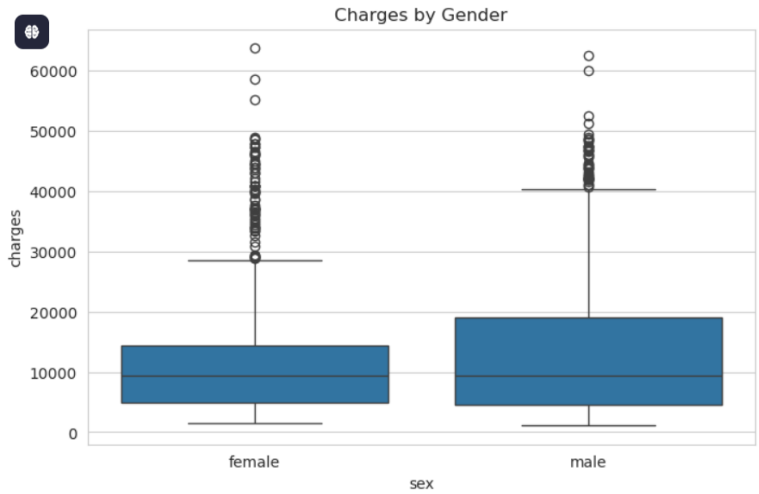
Interpretation:

From this visualization, it is clear that the geographic location of the customer does not have a significant effect on the insurance charges. In other words, the location of the customer (i.e., which region he belongs to) does not have a significant effect on the insurance charges.

Relevance to modelling:

This is relevant to modelling because it is clear that the ‘region’ column does not have a significant effect on the ‘charges’. In other words, the ‘region’ column may not have a high predictive power compared to other columns (like ‘smoking_status’). Hence, this column may have very little effect on the modelling process (James et al., 2021). The EDA process is helpful in ensuring that only relevant columns are emphasized during modeling (Han et al., 2012).

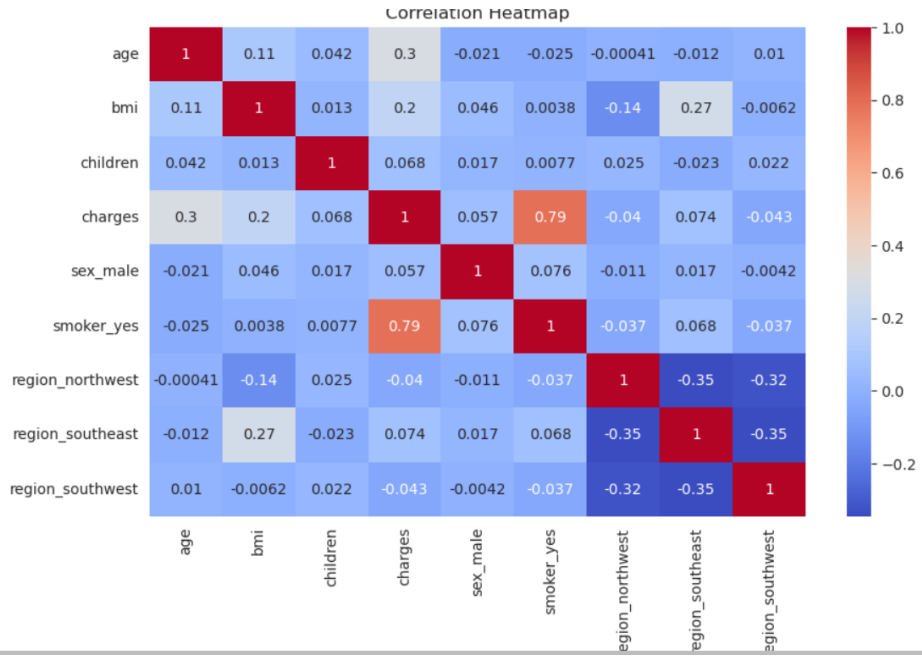
```
[9]: plt.figure(figsize=(8,5))
     sns.boxplot(x='sex', y='charges', data=df)
     plt.title("Charges by Gender")
     plt.show()
```



This screenshot depicts one of the EDA procedures in which the charges for medical insurance are compared for males amd females by means of a boxplot. Before this, the data would have undergone cleaning in which the variable sex would have been correctly labeled as male/female, and charges would have had the correct numerical values.

From the plot, it is evident that the median charges for both males and females are very close. Even though there are outliers in both cases, they are distributed more or less equally. Thus, there does not seem to be any significant difference in charges for males and females.

Interpretation: This implies that gender does not have a significant impact of medical insurance charges. Hence, it is likely to have a very low predictive power in the model. During EDA, such weak relationships are often identified, which can help in improving the performance of the model by concentrating on more impactful variables (James et al., 2021; Han, Kamber & Pei, 2012).

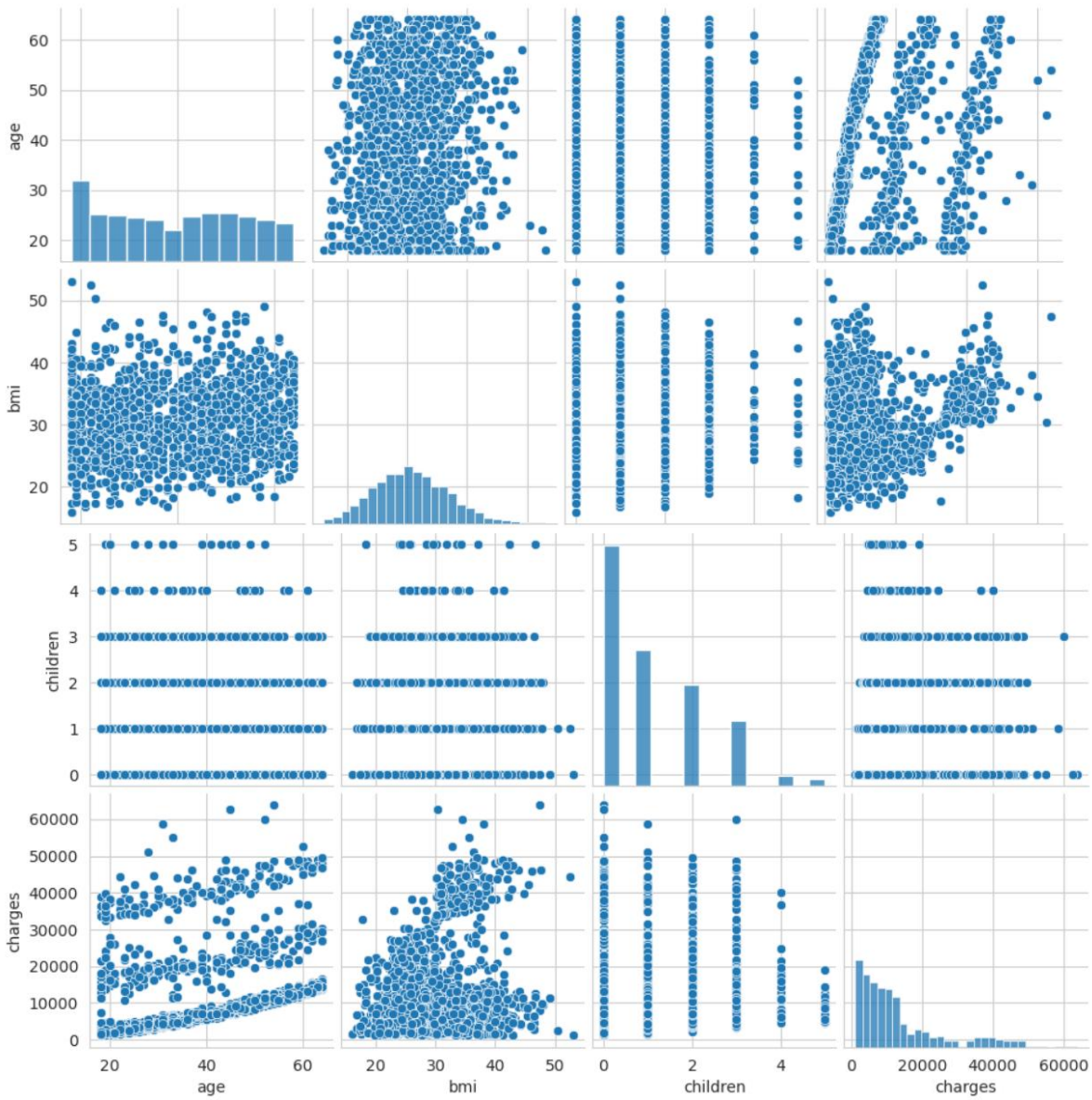


The heatmap is used to represent the correlation between the variables.

Observations:

- Charges are positively correlated with age and BMI.
- Smoking status is the variable that is the most correlated with charges.
- Other variables are not correlated.

Interpretation: This is further evidence that the variables smoking status, age, and BMI are important in determining the insurance charges.

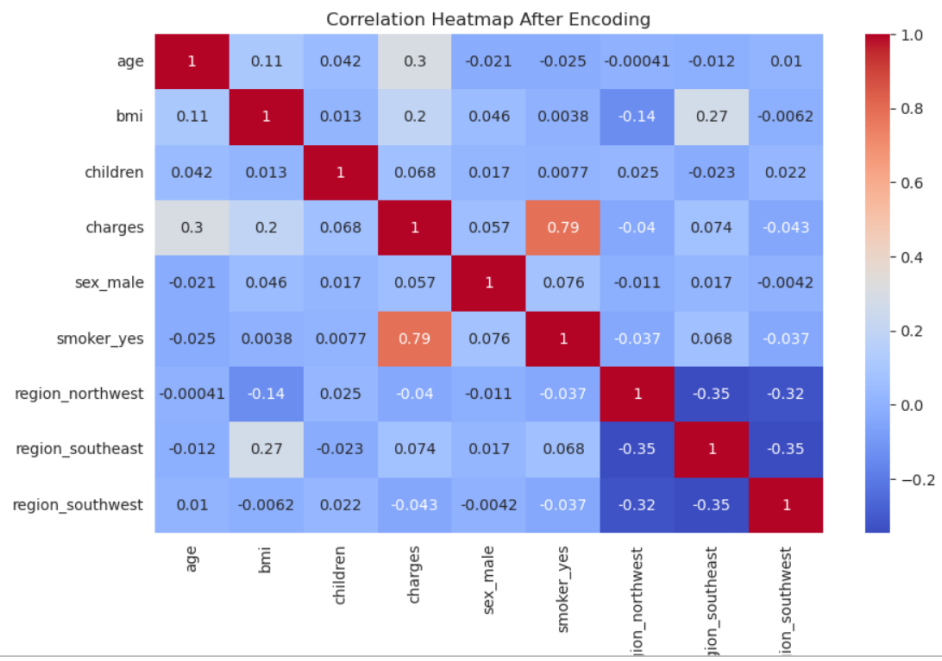


We can get a picture of how several variables are connected to each other using a pairplot.

Observations:

- If we loook at the figure we can see that there is a connection between charges, age, BMI and whether a person smokes or not.
- We can also see that the data is grouped together especially when it comes to people who smoke and those who do not smoke.

Interpretation: The pairplot helps us get an idea of what the data is like. We use the pairplot to look at pairwise relationships and see how variables, like charges and age and BMI and smoking are related to each other.



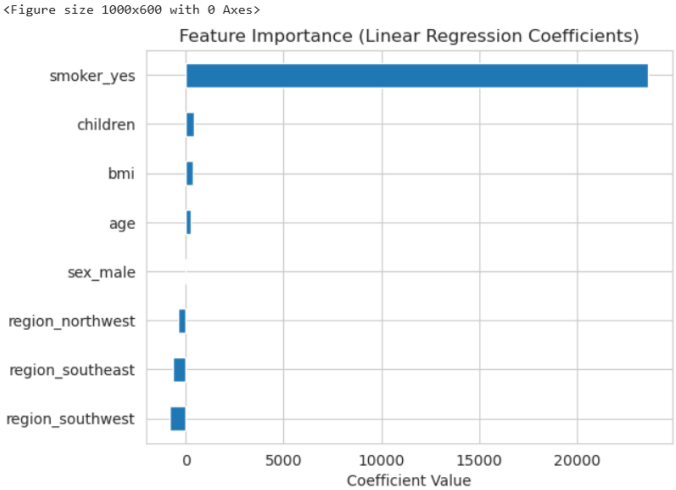
The updated heatmap now includes the relationships between all the numerical features, including the newly encoded features.

- Smoking status continues to be highly correlated with charges
- Age and BMI remain important facotrs

This confirms that the data is now ready for modeling.

[18]:
```
plt.figure(figsize=(10,6))
coefficients.sort_values(by='Coefficient').plot(kind='barh', legend=False)
plt.title("Feature Importance (Linear Regression Coefficients)")
plt.xlabel("Coefficient Value")
plt.show()
```

```
<Figure size 1000x600 with 0 Axes>
```



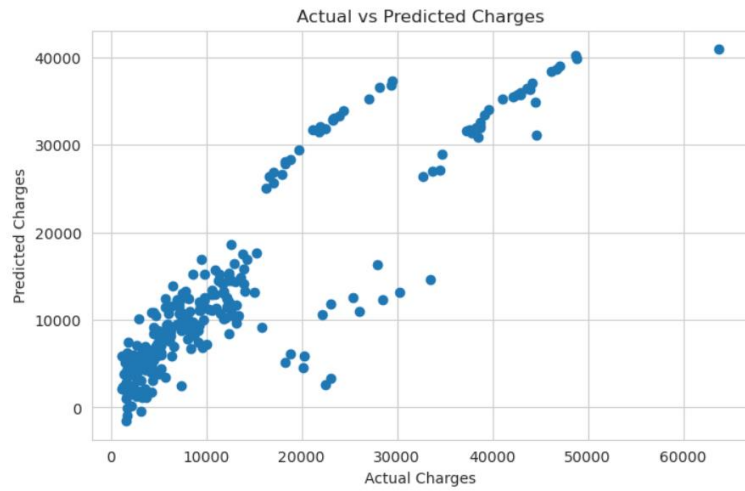
The bar chart above shows the relative importance of each feature in the model.

Observations:

- Features with larger absolute values have a stronger impact on predictions.
- Smoking status is expected to be the most influential feature.

This visualisation makes it easier to interpret the model.

```
[21]: plt.figure(figsize=(8,5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Actual vs Predicted Charges")
plt.show()
```



When we look at the scatter plot we see the charges and the predicted charges side by side.

Observations:

The points that are close to the line show that the predicted charges were pretty much the same as the actual charges

We can see that some points are not close to the line, which means the model made some mistakes

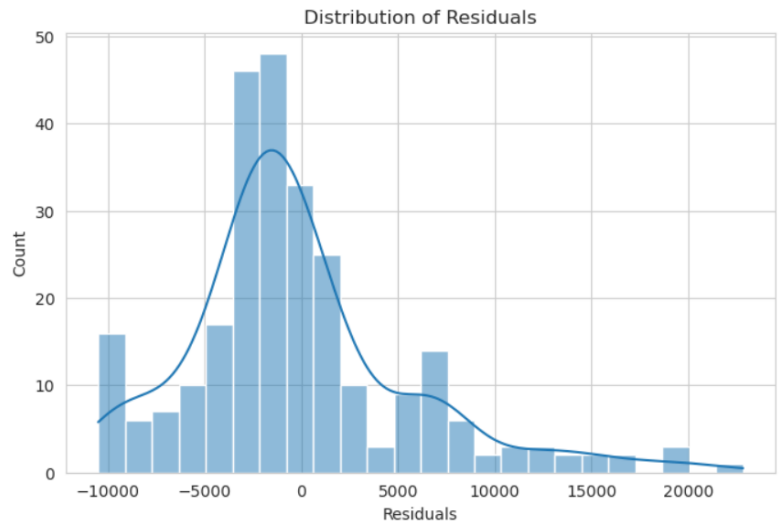
The model is doing a good job overall because we can see an upward trend in the points

Interpretation:

The model is doing a good job with the predicted values but we can still see some mistakes, which means the model needs a little more work to get better. The actual charges and the predicted charges are mostly in line, which's a good sign, for the model.

```
residuals = y_test - y_pred

plt.figure(figsize=(8,5))
sns.histplot(residuals, kde=True)
plt.title("Distribution of Residuals")
plt.xlabel("Residuals")
plt.show()
```



Residuals are the difference between what happened and what the Residual Analysis predicted would happen.

Observations:

When we look at the Residual Analysis the way the residuals are spread out tells us how good the model is at making predictions

The residuals should be evenly spread out around zero for the Residual Analysis to be

Interpretation:

If the Residual Analysis shows that the residuals are centered around zero that means the model predictions are pretty accurate and do not favor one side.. If the Residual Analysis shows that the residuals are not evenly spread out or if they are skewed, to one side that means the Residual Analysis has some problems and we need to make the model better. The Residual Analysis is very important to check how good the model is.

Reference List

James, G., Witten, D., Hastie, T. and Tibshirani, R., 2021. An Introduction to Statistical Learning. 2nd ed. New York: Springer.

Han, J., Kamber, M. and Pei, J., 2012. Data Mining: Concepts and Techniques. 3rd ed. Waltham: Morgan Kaufmann.

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Géron, A., 2019. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. 2nd ed. Sebastopol: O'Reilly Media.